

Milestone 7: Priced Scenarios and Avoided Cost

Literature-anchored lift, break-even, and a separate avoided-medical-cost arm

Prepared by: KSU mHealth Research Lab | Date: 2026-10-02

RESEARCH FRAMEWORK. Lift is research-generated or taken from published trials; no InComm lift, financial or issuance data is used. Dollar values are research ranges tagged by provenance.

1. Priced ROI under published incentive-trial effects

Relative risks of closure: 1.00 (no effect; the IDEAS diabetic eye screening RCT), 1.17 (colorectal screening, 8 RCTs) and 1.42 (pooled screening meta-analysis). Carried as odds ratios so probabilities stay within 0 to 1. Offer to all, pay on completion. These are scenarios, not InComm results.

Rr Case	Odds Ratio	Value Case	Mean Implied Rr	Net Total Usd	Cost Total Usd	ROI
base	1.2619	base	1.1489	-24,733	110,752	-0.2233
base	1.2619	high	1.1489	31,641	110,752	0.2857
base	1.2619	low	1.1489	-54,841	110,752	-0.4952
high	1.7317	base	1.3665	86,553	128,563	0.6732
high	1.7317	high	1.3665	227,504	128,563	1.7696
high	1.7317	low	1.3665	11,261	128,563	0.0876
low	1.0000	base	1.0000	-98,224	98,224	-1.0000
low	1.0000	high	1.0000	-98,224	98,224	-1.0000
low	1.0000	low	1.0000	-98,224	98,224	-1.0000

2. Break-even relative risk by gap (Stars value only)

Care Gap Code	Value Case	Break Even Rr
BH_POST_DISCHARGE_7D	base	n/a
BH_POST_DISCHARGE_7D	high	n/a
BH_POST_DISCHARGE_7D	low	n/a
COPD_SPIROMETRY	base	n/a
COPD_SPIROMETRY	high	n/a
COPD_SPIROMETRY	low	n/a
DIAB_A1C_TEST	base	1.0826

DIAB_A1C_TEST	high	1.0482
DIAB_A1C_TEST	low	1.1369
DIAB_BLOOD_SUGAR	base	1.0826
DIAB_BLOOD_SUGAR	high	1.0482
DIAB_BLOOD_SUGAR	low	1.1369
DIAB_EYE_EXAM	base	1.1794
DIAB_EYE_EXAM	high	1.1004
DIAB_EYE_EXAM	low	1.3032
DIAB_NEPHRO_SCREEN	base	1.3216
DIAB_NEPHRO_SCREEN	high	1.1607
DIAB_NEPHRO_SCREEN	low	1.6556
HTN_PCP_FOLLOWUP	base	1.1021
HTN_PCP_FOLLOWUP	high	1.0618
HTN_PCP_FOLLOWUP	low	1.1512

3. Avoided medical cost arm (separate; never summed into the Stars headline)

- Only BH 7-day follow-up has a published closure-to-readmission link. It is observational (adjusted odds ratios 0.88 to 0.91), so the low case is zero effect.
- Every other gap is NOT_MODELLED: absence of verified evidence, not proof of no benefit. Screening often raises near-term spending, so no value is guessed.
- One 30-day event, no discounting. The plan bears the cost under capitation.
- Inputs: readmission rate 11 / 20 / 36% (CMS inpatient psychiatric facility data); odds reduction 0 / 0.10 / 0.22; cost per readmission \$8,800 / \$12,150 / \$15,500 (HCUP, Medicare).

Care Gap Code	Value Case	Closure Value Usd	Avoided Cost Usd	Net Per Offer Usd	Net Per Offer With Avoided Usd	Break Even Rr With Avoided
BH_POS T_DISCH ARGE_7 D	low	0	1.221e-13	-17.4528	-17.4528	n/a
BH_POS T_DISCH ARGE_7 D	base	0	198.3673	-17.4528	-9.1981	1.1259
BH_POS T_DISCH ARGE_7 D	high	0	853.2407	-17.4528	18.0534	1.0267

Reading: at base, BH follow-up avoids \$198 per closure and still nets \$-9.20 per offer under the base literature lift (implied RR 1.06 against a break-even of 1.13). It pays only in the high case. BH baseline closure is already 74.5%, so there is little room to lift.